P03

Acoustic Resonance and Energy Dissipation in Coupled Cavities

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Domain DOM_PHYS

Abstract

Helmholtz resonators demonstrate sharp acoustic impedance minima at resonance. The quality factor Q measures the bandwidth of peak acoustic absorption relative to viscous boundary layer dissipation.

Abstract quoted from U. Ingard et al., Journal of the Acoustical Society of America (1953).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?